

Q vs Q - Stat Tests

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1 Imports

```
[45]: import numpy as np
import pandas as pd
from scipy.stats import kruskal
import scikit_posthocs as sp
```

2 Logit, Firm 1

2.1 Data Loading

```
[46]: model = 'logit' # row number = 2

avg_prices = {
    "shock_none": [],
    "shock_a": [],
    "shock_b": [],
    "shock_c": []
}

deltas = {
    "shock_none": [],
    "shock_a": [],
    "shock_b": [],
    "shock_c": []
}

rpdis = {
    "shock_none": [],
    "shock_a": [],
    "shock_b": [],
    "shock_c": []
}

for i in range(1, 8):
    # load the dataframes
    df_shock_none = pd.read_csv(f"../results/run_{i}/shock_none/q_vs_q.csv")
```

```

df_shock_a = pd.read_csv(f"../results/run_{i}/shock_a/q_vs_q_schemeA.csv")
df_shock_b = pd.read_csv(f"../results/run_{i}/shock_b/q_vs_q_schemeB.csv")
df_shock_c = pd.read_csv(f"../results/run_{i}/shock_c/q_vs_q_schemeC.csv")

# average prices
avg_prices["shock_none"].append(df_shock_none.iat[0, 1])
avg_prices["shock_a"].append(df_shock_a.iat[0, 1])
avg_prices["shock_b"].append(df_shock_b.iat[0, 1])
avg_prices["shock_c"].append(df_shock_c.iat[0, 1])

# deltas
deltas["shock_none"].append(df_shock_none.iat[0, 6])
deltas["shock_a"].append(df_shock_a.iat[0, 6])
deltas["shock_b"].append(df_shock_b.iat[0, 6])
deltas["shock_c"].append(df_shock_c.iat[0, 6])

# rpdis
rpdis["shock_none"].append(df_shock_none.iat[0, 8])
rpdis["shock_a"].append(df_shock_a.iat[0, 8])
rpdis["shock_b"].append(df_shock_b.iat[0, 8])
rpdis["shock_c"].append(df_shock_c.iat[0, 8])

```

2.2 Kruskal-Wallis Test - Average Prices

```

[47]: h_stat, p_val = kruskal(
    avg_prices["shock_none"],
    avg_prices["shock_a"],
    avg_prices["shock_b"],
    avg_prices["shock_c"]
)
n = 4 * 7 # total number of observations across all groups
epsilon_squared = h_stat / (n - 1)

print("Kruskal-Wallis Test on Average Prices")
print(f"H-statistic: {h_stat:.2f}, P-value: {p_val:.8f}, epsilon_squared: {epsilon_squared:.4f}\n")

```

Kruskal-Wallis Test on Average Prices

H-statistic: 23.68, P-value: 0.00002918, epsilon_squared: 0.8769

2.3 Post-Hoc Testing - Average Prices

```

[48]: dunn_matrix = sp.posthoc_dunn(
    [
        avg_prices["shock_none"],
        avg_prices["shock_a"],

```

```

        avg_prices["shock_b"],
        avg_prices["shock_c"]
    ], p_adjust='holm'
)

print(dunn_matrix)

```

	1	2	3	4
1	1.000000	0.006367	1.000000	0.00138
2	0.006367	1.000000	0.006367	1.00000
3	1.000000	0.006367	1.000000	0.00138
4	0.001380	1.000000	0.001380	1.00000

Wherever a row and a column intersect with a value < 0.05 , those two specific groups are statistically significantly different from each other. As seen from the `dunn_matrix`,

- `avg_prices["shock_none"]` is significantly different from `avg_prices["shock_a"]`
- `avg_prices["shock_none"]` is significantly different from `avg_prices["shock_c"]`
- `avg_prices["shock_a"]` is significantly different from `avg_prices["shock_b"]`
- `avg_prices["shock_b"]` is significantly different from `avg_prices["shock_c"]`

2.4 Kruskal-Wallis Test - Deltas

```

[49]: h_stat, p_val = kruskal(
        deltas["shock_none"],
        deltas["shock_a"],
        deltas["shock_b"],
        deltas["shock_c"]
    )
    n = 4 * 7 # total number of observations across all groups
    epsilon_squared = h_stat / (n - 1)

    print("Kruskal-Wallis Test on Deltas")
    print(f"H-statistic: {h_stat:.2f}, P-value: {p_val:.8f}, epsilon_squared: {epsilon_squared:.4f}\n")

```

Kruskal-Wallis Test on Deltas

H-statistic: 25.52, P-value: 0.00001202, epsilon_squared: 0.9452

2.5 Post-Hoc Testing - Deltas

```

[50]: dunn_matrix = sp.posthoc_dunn(
        [
            deltas["shock_none"],
            deltas["shock_a"],
            deltas["shock_b"],
            deltas["shock_c"]
        ], p_adjust='holm'
    )

```

```
)

print(dunn_matrix)
```

```

      1      2      3      4
1  1.000000  0.006993  0.330476  0.000010
2  0.006993  1.000000  0.330476  0.330476
3  0.330476  0.330476  1.000000  0.006993
4  0.000010  0.330476  0.006993  1.000000
```

- `deltas["shock_none"]` is significantly different from `deltas["shock_a"]`
- `deltas["shock_none"]` is significantly different from `deltas["shock_c"]`
- `deltas["shock_b"]` is significantly different from `deltas["shock_c"]`

2.6 Kruskal-Wallis Test - RPDIs

```
[51]: h_stat, p_val = kruskal(
        rpdis["shock_none"],
        rpdis["shock_a"],
        rpdis["shock_b"],
        rpdis["shock_c"]
    )
    n = 4 * 7 # total number of observations across all groups
    epsilon_squared = h_stat / (n - 1)

    print("Kruskal-Wallis Test on RPDIs")
    print(f"H-statistic: {h_stat:.2f}, P-value: {p_val:.8f}, epsilon_squared: {epsilon_squared:.4f}\n")
```

Kruskal-Wallis Test on RPDIs

H-statistic: 25.75, P-value: 0.00001074, epsilon_squared: 0.9538

2.7 Post-Hoc Testing - RPDIs

```
[52]: dunn_matrix = sp.posthoc_dunn(
        [
            rpdis["shock_none"],
            rpdis["shock_a"],
            rpdis["shock_b"],
            rpdis["shock_c"]
        ], p_adjust='holm'
    )

    print(dunn_matrix)
```

```

      1      2      3      4
1  1.000000  0.006646  0.325618  0.000009
2  0.006646  1.000000  0.325618  0.325618
```

```

3  0.325618  0.325618  1.000000  0.006646
4  0.000009  0.325618  0.006646  1.000000

```

- `rpdis["shock_none"]` is significantly different from `rpdis["shock_a"]`
- `rpdis["shock_none"]` is significantly different from `rpdis["shock_c"]`
- `rpdis["shock_b"]` is significantly different from `rpdis["shock_c"]`

3 Hotelling, Firm 1

3.1 Data Loading

```

[53]: model = 'hotelling' # row number = 2

avg_prices = {
    "shock_none": [],
    "shock_a": [],
    "shock_b": [],
    "shock_c": []
}

deltas = {
    "shock_none": [],
    "shock_a": [],
    "shock_b": [],
    "shock_c": []
}

rpdis = {
    "shock_none": [],
    "shock_a": [],
    "shock_b": [],
    "shock_c": []
}

for i in range(1, 8):
    # load the dataframes
    df_shock_none = pd.read_csv(f"../results/run_{i}/shock_none/q_vs_q.csv")
    df_shock_a = pd.read_csv(f"../results/run_{i}/shock_a/q_vs_q_schemeA.csv")
    df_shock_b = pd.read_csv(f"../results/run_{i}/shock_b/q_vs_q_schemeB.csv")
    df_shock_c = pd.read_csv(f"../results/run_{i}/shock_c/q_vs_q_schemeC.csv")

    # average prices
    avg_prices["shock_none"].append(df_shock_none.iat[1, 1])
    avg_prices["shock_a"].append(df_shock_a.iat[1, 1])
    avg_prices["shock_b"].append(df_shock_b.iat[1, 1])
    avg_prices["shock_c"].append(df_shock_c.iat[1, 1])

    # deltas

```

```

deltas["shock_none"].append(df_shock_none.iat[1, 6])
deltas["shock_a"].append(df_shock_a.iat[1, 6])
deltas["shock_b"].append(df_shock_b.iat[1, 6])
deltas["shock_c"].append(df_shock_c.iat[1, 6])

# rpdís
rpdís["shock_none"].append(df_shock_none.iat[1, 8])
rpdís["shock_a"].append(df_shock_a.iat[1, 8])
rpdís["shock_b"].append(df_shock_b.iat[1, 8])
rpdís["shock_c"].append(df_shock_c.iat[1, 8])

```

3.2 Kruskal-Wallis Test - Average Prices

```

[54]: h_stat, p_val = kruskal(
    avg_prices["shock_none"],
    avg_prices["shock_a"],
    avg_prices["shock_b"],
    avg_prices["shock_c"]
)
n = 4 * 7 # total number of observations across all groups
epsilon_squared = h_stat / (n - 1)

print("Kruskal-Wallis Test on Average Prices")
print(f"H-statistic: {h_stat:.2f}, P-value: {p_val:.8f}, epsilon_squared: {epsilon_squared:.4f}\n")

```

Kruskal-Wallis Test on Average Prices

H-statistic: 27.00, P-value: 0.00000589, epsilon_squared: 1.0000

3.3 Post-Hoc Testing - Average Prices

```

[55]: dunn_matrix = sp.posthoc_dunn(
    [
        avg_prices["shock_none"],
        avg_prices["shock_a"],
        avg_prices["shock_b"],
        avg_prices["shock_c"]
    ], p_adjust='holm'
)

print(dunn_matrix)

```

	1	2	3	4
1	1.000000	0.001431	1.000000	0.001431
2	0.001431	1.000000	0.001431	1.000000
3	1.000000	0.001431	1.000000	0.001431
4	0.001431	1.000000	0.001431	1.000000

- `avg_prices["shock_none"]` is significantly different from `avg_prices["shock_a"]`
- `avg_prices["shock_none"]` is significantly different from `avg_prices["shock_c"]`
- `avg_prices["shock_a"]` is significantly different from `avg_prices["shock_b"]`
- `avg_prices["shock_b"]` is significantly different from `avg_prices["shock_c"]`

3.4 Kruskal-Wallis Test - Deltas

```
[56]: h_stat, p_val = kruskal(
        deltas["shock_none"],
        deltas["shock_a"],
        deltas["shock_b"],
        deltas["shock_c"]
    )
    n = 4 * 7 # total number of observations across all groups
    epsilon_squared = h_stat / (n - 1)

    print("Kruskal-Wallis Test on Deltas")
    print(f"H-statistic: {h_stat:.2f}, P-value: {p_val:.8f}, epsilon_squared: {epsilon_squared:.4f}\n")
```

Kruskal-Wallis Test on Deltas

H-statistic: 7.56, P-value: 0.05595521, epsilon_squared: 0.2801

As the P-value is not significant (> 0.05), there is no significant difference between the deltas

3.5 Post-Hoc Testing - Deltas

If the Kruskal-Wallis P-value is not significant, the Dunn matrix is not worth considering.

3.6 Kruskal-Wallis Test - RPDIs

```
[57]: h_stat, p_val = kruskal(
        rpdis["shock_none"],
        rpdis["shock_a"],
        rpdis["shock_b"],
        rpdis["shock_c"]
    )
    n = 4 * 7 # total number of observations across all groups
    epsilon_squared = h_stat / (n - 1)

    print("Kruskal-Wallis Test on RPDIs")
    print(f"H-statistic: {h_stat:.2f}, P-value: {p_val:.8f}, epsilon_squared: {epsilon_squared:.4f}\n")
```

Kruskal-Wallis Test on RPDIs

H-statistic: 23.21, P-value: 0.00003643, epsilon_squared: 0.8598

3.7 Post-Hoc Testing - RPDIs

```
[58]: dunn_matrix = sp.posthoc_dunn(
      [
        rpdis["shock_none"],
        rpdis["shock_a"],
        rpdis["shock_b"],
        rpdis["shock_c"]
      ], p_adjust='holm'
    )

print(dunn_matrix)
```

	1	2	3	4
1	1.000000	0.003117	0.690564	0.000219
2	0.003117	1.000000	0.039700	0.690564
3	0.690564	0.039700	1.000000	0.005785
4	0.000219	0.690564	0.005785	1.000000

- rpdis["shock_none"] is significantly different from rpdis["shock_a"]
- rpdis["shock_none"] is significantly different from rpdis["shock_c"]
- rpdis["shock_a"] is significantly different from rpdis["shock_b"]
- rpdis["shock_b"] is significantly different from rpdis["shock_c"]

4 Linear, Firm 1

4.1 Data Loading

```
[59]: model = 'linear' # row number = 2

avg_prices = {
    "shock_none": [],
    "shock_a": [],
    "shock_b": [],
    "shock_c": []
}

deltas = {
    "shock_none": [],
    "shock_a": [],
    "shock_b": [],
    "shock_c": []
}

rpdis = {
    "shock_none": [],
    "shock_a": [],
    "shock_b": [],
```



```

    "shock_c": []
}

for i in range(1, 8):
    # load the dataframes
    df_shock_none = pd.read_csv(f"../results/run_{i}/shock_none/q_vs_q.csv")
    df_shock_a = pd.read_csv(f"../results/run_{i}/shock_a/q_vs_q_schemeA.csv")
    df_shock_b = pd.read_csv(f"../results/run_{i}/shock_b/q_vs_q_schemeB.csv")
    df_shock_c = pd.read_csv(f"../results/run_{i}/shock_c/q_vs_q_schemeC.csv")

    # average prices
    avg_prices["shock_none"].append(df_shock_none.iat[2, 1])
    avg_prices["shock_a"].append(df_shock_a.iat[2, 1])
    avg_prices["shock_b"].append(df_shock_b.iat[2, 1])
    avg_prices["shock_c"].append(df_shock_c.iat[2, 1])

    # deltas
    deltas["shock_none"].append(df_shock_none.iat[2, 6])
    deltas["shock_a"].append(df_shock_a.iat[2, 6])
    deltas["shock_b"].append(df_shock_b.iat[2, 6])
    deltas["shock_c"].append(df_shock_c.iat[2, 6])

    # rpdis
    rpdis["shock_none"].append(df_shock_none.iat[2, 8])
    rpdis["shock_a"].append(df_shock_a.iat[2, 8])
    rpdis["shock_b"].append(df_shock_b.iat[2, 8])
    rpdis["shock_c"].append(df_shock_c.iat[2, 8])

```

4.2 Kruskal-Wallis Test - Average Prices

```

[60]: h_stat, p_val = kruskal(
    avg_prices["shock_none"],
    avg_prices["shock_a"],
    avg_prices["shock_b"],
    avg_prices["shock_c"]
)
n = 4 * 7 # total number of observations across all groups
epsilon_squared = h_stat / (n - 1)

print("Kruskal-Wallis Test on Average Prices")
print(f"H-statistic: {h_stat:.2f}, P-value: {p_val:.8f}, epsilon_squared: {epsilon_squared:.4f}\n")

```

Kruskal-Wallis Test on Average Prices

H-statistic: 15.00, P-value: 0.00181665, epsilon_squared: 0.5556

4.3 Post-Hoc Testing - Average Prices

```
[61]: dunn_matrix = sp.posthoc_dunn(
      [
        avg_prices["shock_none"],
        avg_prices["shock_a"],
        avg_prices["shock_b"],
        avg_prices["shock_c"]
      ], p_adjust='holm'
    )

print(dunn_matrix)
```

	1	2	3	4
1	1.000000	0.037019	1.000000	0.037019
2	0.037019	1.000000	0.037019	1.000000
3	1.000000	0.037019	1.000000	0.037019
4	0.037019	1.000000	0.037019	1.000000

- avg_prices["shock_none"] is significantly different from avg_prices["shock_a"]
- avg_prices["shock_none"] is significantly different from avg_prices["shock_c"]
- avg_prices["shock_a"] is significantly different from avg_prices["shock_b"]
- avg_prices["shock_b"] is significantly different from avg_prices["shock_c"]

4.4 Kruskal-Wallis Test - Deltas

```
[62]: h_stat, p_val = kruskal(
      deltas["shock_none"],
      deltas["shock_a"],
      deltas["shock_b"],
      deltas["shock_c"]
    )
n = 4 * 7 # total number of observations across all groups
epsilon_squared = h_stat / (n - 1)

print("Kruskal-Wallis Test on Deltas")
print(f"H-statistic: {h_stat:.2f}, P-value: {p_val:.8f}, epsilon_squared: {epsilon_squared:.4f}\n")
```

Kruskal-Wallis Test on Deltas

H-statistic: 23.58, P-value: 0.00003059, epsilon_squared: 0.8733

4.5 Post-Hoc Testing - Deltas

```
[63]: dunn_matrix = sp.posthoc_dunn(
      [
        deltas["shock_none"],
        deltas["shock_a"],
```

```

        deltas["shock_b"],
        deltas["shock_c"]
    ], p_adjust='holm'
)

print(dunn_matrix)

```

	1	2	3	4
1	1.000000	0.086278	0.645249	0.000729
2	0.086278	1.000000	0.032470	0.214227
3	0.645249	0.032470	1.000000	0.000124
4	0.000729	0.214227	0.000124	1.000000

- `deltas["shock_none"]` is significantly different from `deltas["shock_c"]`
- `deltas["shock_a"]` is significantly different from `deltas["shock_b"]`
- `deltas["shock_b"]` is significantly different from `deltas["shock_c"]`

4.6 Kruskal-Wallis Test - RPDIs

```

[64]: h_stat, p_val = kruskal(
        rpd["shock_none"],
        rpd["shock_a"],
        rpd["shock_b"],
        rpd["shock_c"]
    )
    n = 4 * 7 # total number of observations across all groups
    epsilon_squared = h_stat / (n - 1)

    print("Kruskal-Wallis Test on RPDIs")
    print(f"H-statistic: {h_stat:.2f}, P-value: {p_val:.8f}, epsilon_squared: {epsilon_squared:.4f}\n")

```

Kruskal-Wallis Test on RPDIs

H-statistic: 22.97, P-value: 0.00004103, epsilon_squared: 0.8506

4.7 Post-Hoc Testing - RPDIs

```

[65]: dunn_matrix = sp.posthoc_dunn(
        [
            rpd["shock_none"],
            rpd["shock_a"],
            rpd["shock_b"],
            rpd["shock_c"]
        ], p_adjust='holm'
    )

    print(dunn_matrix)

```

	1	2	3	4
1	1.000000	0.004607	0.955365	0.000344
2	0.004607	1.000000	0.027686	0.955365
3	0.955365	0.027686	1.000000	0.004607
4	0.000344	0.955365	0.004607	1.000000

- `rpdis["shock_none"]` is significantly different from `rpdis["shock_a"]`
- `rpdis["shock_none"]` is significantly different from `rpdis["shock_c"]`
- `rpdis["shock_a"]` is significantly different from `rpdis["shock_b"]`
- `rpdis["shock_b"]` is significantly different from `rpdis["shock_c"]`